

HITL Setup for TARLAND

1. Switch to the branch `subash/tarland`
2. Make sure the ardupilot development environment is setup as outlined in [building setup for linux](#)
3. Update the following line in the file `ardupilot/Tools/scripts/sitl-on-hardware/extra-hwdef-sitl-on-hw.dat` `define MODE_FOLLOW_ENABLED 1`
4. Connect the `CubeOrangePlus` and build the code using following command

```
cd ~/ardupilot
./Tools/scripts/sitl-on-hardware/sitl-on-hw.py --board
CubeOrangePlus-SimOnHardWare --vehicle copter --simclass
MultiCopter --frame X --upload
```

5. Find the device id by running the following command

```
ls /dev/serial/by-id/
```

It will list the device id as follows

```
usb-CubePilot_CubeOrange+_3E0021000C51333130373938-if00
usb-CubePilot_CubeOrange+_3E0021000C51333130373938-if02
```

5. Once the build is complete connect to MAVProxy with the following command

```
mavproxy.py --master=/dev/serial/by-id/usb-
CubePilot_CubeOrange+_3E0021000C51333130373938-if00 --map --console
```

6. Load the `mav_tarland.parm` file to the board using mavproxy or using GCS

```
param load <path to mav_tarland.parm file>
param save
```

7. Now switch to Guided mode, take off to a height and then switch to mode tarland (mode 29)

```
mode GUIDED
```

```
takeoff 15
```

```
mode 29
```

8. It will switch to Mode Tarland. For this mode to work properly, simulate a GPS message of the target and send it to `GLOBAL_POSITION_INT` mavlink message.